

Exponential equations

Get the same base on both sides, then equate the indices — and know the three tricks for when you cannot.

LESSON 51–52

2 × 40 MIN

ALGEBRA 10, §3.2

P2 · 2.3

LESSON CLOCK

10'

22'

24'

22'

20'

12'

The one-to-one property	10 min
A common base	22 min
Substitution	24 min
Homogeneous equations	22 min
Practice	20 min
Homework	12 min

LEARNING OBJECTIVES

- Solve $a^{f(x)} = a^{g(x)}$ by equating indices.
- Rewrite both sides to a common base.
- Solve equations that become quadratic after a substitution.
- Solve homogeneous equations by dividing through.

TEXTBOOK

National	pp. 197–208
Cambridge	Pure Mathematics 2 & 3, pp. 33–38
Workbook	P2 Exercise 2B

01

Explanation

The method rests on one property

A^x IS ONE-TO-ONE

Because $y = a^x$ is strictly monotonic, $a^m = a^n$ forces $m = n$ (for $a > 0, a \neq 1$). That single fact solves most exponential equations.

$$2^{(3x-1)} = 2^{(x+5)} \Rightarrow 3x - 1 = x + 5 \Rightarrow x = 3$$

So the whole task is: **make the bases the same.**

Getting a common base

EQUATION	REWRITE AS	SOLUTION
$2^x = 32$	$2^x = 2^5$	$x = 5$
$3^{(2x)} = 81$	$3^{(2x)} = 3^4$	$x = 2$
$4^x = 8$	$2^{(2x)} = 2^3$	$x = 1.5$
$(\frac{1}{3})^x = 9$	$3^{(-x)} = 3^2$	$x = -2$
$5^{(x^2-3x)} = 1$	$5^{(x^2-3x)} = 5^0$	$x = 0, 3$

EVERY NUMBER IS A POWER OF SOMETHING

$32 = 2^5$, $81 = 3^4$, $8 = 2^3$, $0.25 = 2^{-2}$, $1 = a^0$. Learning the powers of 2, 3, 5 up to a few hundred removes most of the difficulty in this topic.

Substitution

When $a^{(2x)}$ and a^x both appear, set $t = a^x$ — and remember $t > 0$:

$$4^x - 5 \cdot 2^x + 4 = 0, t = 2^x > 0 \Rightarrow t^2 - 5t + 4 = 0$$

$t = 1$ or $t = 4$; both positive, so $2^x = 1$ gives $x = 0$ and $2^x = 4$ gives $x = 2$.

WRITE $T > 0$ WITH THE SUBSTITUTION

A negative t is rejected immediately, because a^x is never negative. That single line saves the whole check at the end — and it is the mark the examiner is looking for.

Note $4^x = (2^2)^x = (2^x)^2 = t^2$: recognising that is the step, not the algebra after it.

Homogeneous equations

When every term has the same total degree in two bases, divide through by one of them:

$$9^x - 4 \cdot 6^x + 3 \cdot 4^x = 0$$

Divide by 4^x (never zero) to get, with $t = (\frac{3}{2})^x > 0$:

$$t^2 - 4t + 3 = 0 \Rightarrow t = 1, 3 \Rightarrow x = 0, x = \log_{(1.5)} 3$$

The pattern to spot: $9^x = (3^2)^x$, $6^x = (2 \cdot 3)^x$, $4^x = (2^2)^x$ — the bases 9, 6, 4 are 3^2 , $3 \cdot 2$, 2^2 , a perfect quadratic pattern.

02 Worked examples

EXAMPLE 1 Solve $2^{x+3} = 32$.

① $32 = 2^5$

② $x + 3 = 5$

③ $x = 2$

ANSWER

$x = 2$

EXAMPLE 2 Solve $9^x - 10 \cdot 3^x + 9 = 0$.

① Let $t = 3^x > 0$; $9^x = t^2$.

② $t^2 - 10t + 9 = 0$

③ $t = 1$ or $t = 9$, both positive.

④ $3^x = 1 \Rightarrow x = 0$; $3^x = 9 \Rightarrow x = 2$.

ANSWER

$x = 0$ and $x = 2$

EXAMPLE 3 Solve $2^{x+1} + 2^{x-1} = 20$.

$$① \quad 2 \cdot 2^x + \frac{2^x}{2} = 20$$

Factor out 2^x .

$$② \quad 2^x(2 + 0.5) = 20$$

$$③ \quad 2^x = 8$$

$$④ \quad x = 3$$

ANSWER

$$x = 3$$

03 **Interactive model**

Write the powers of 2 and 3 on the board and leave them there for the whole lesson.

Powers and their bases

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**index n **3**power m **1**check: value³ **64**a¹ **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 **Terminology**

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Exponential equation	Ko'rsatkichli tenglama	Показательное уравнение
Common base	Umumiy asos	Общее основание
Equating indices	Ko'rsatkichlarni tenglashtirish	Приравнивание показателей
Substitution	Almashtirish	Замена
Homogeneous equation	Bir jinsli tenglama	Однородное уравнение
Auxiliary variable	Yordamchi o'zgaruvchi	Вспомогательная переменная
Positivity condition	Musbatlik sharti	Условие положительности
Rejected root	Rad etilgan ildiz	Отброшенный корень

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$2^x = 64$ gives:

$a^m = a^n$ implies $m = n$ because a^x is:

For $4^x - 5 \cdot 2^x + 4 = 0$ substitute:

The condition on that t is:

$$t \geq 0$$

$$t > 0$$

$$t \neq 0$$

none

$5^{(x^2-4)} = 1$ gives:

$$x = 4$$

$$x = \pm 2$$

$$x = 0$$

no solution

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $2^x = 16$

ANSWER $x = 4$

2. Solve $3^x = 27$

ANSWER $x = 3$

3. Solve $5^x = 1$

ANSWER $x = 0$

4. Solve $2^{(x+1)} = 8$

ANSWER $x = 2$

5. Solve $10^x = 0.01$

ANSWER $x = -2$

6. Solve $4^x = 2$

ANSWER $x = 0.5$

7. Solve $(\frac{1}{2})^x = 8$

ANSWER $x = -3$

Medium · the standard question

7 PROBLEMS

1. Solve $2^{3x-1} = 2^{x+5}$

ANSWER $x = 3$

2. Solve $9^x = 27$

ANSWER $x = 1.5$

3. Solve $5^{(x^2-3x)} = 1$

ANSWER $x = 0, 3$

4. Solve $4^x - 5 \cdot 2^x + 4 = 0$

ANSWER $x = 0, 2$

5. Solve $9^x - 10 \cdot 3^x + 9 = 0$

ANSWER $x = 0, 2$

6. Solve $2^{(x+1)} + 2^{(x-1)} = 20$

ANSWER $x = 3$

7. Solve $3^{(x+2)} = 9^{(x-1)}$

ANSWER $x = 4$

Hard · stretch and reason

7 PROBLEMS

1. Solve $4^x - 3 \cdot 2^x - 4 = 0$

ANSWER $x = 2$

2. Solve $9^x - 4 \cdot 6^x + 3 \cdot 4^x = 0$

ANSWER $x = 0$ and $x = \log_{1.5} 3$

3. Solve $2^{2x} - 6 \cdot 2^x + 8 = 0$

ANSWER $x = 1, 2$

4. Solve $3^{x+1} + 3^{x-1} = 30$

ANSWER $x = 2$

5. Solve $25^x - 6 \cdot 5^x + 5 = 0$

ANSWER $x = 0, 1$

6. Solve $2^{x^2} = 4^x$

ANSWER $x = 0, 2$

7. Explain why $4^x + 2^x + 1 = 0$ has no solution

ANSWER Every term is positive

07 Homework

Homework — 6 tasks

Write $t > 0$ with every substitution.

1. Solve $3^{2x-1} = 81$ and $8^x = 32$.

2. Solve $2^{x^2-5x} = 1$.

3. Solve $9^x - 4 \cdot 3^x + 3 = 0$.

4. Solve $5^{x+1} + 5^{x-1} = 130$.

5. Solve $4^x - 6 \cdot 2^x + 8 = 0$.

6. Explain in three sentences why the substitution $t = a^x$ must carry the condition $t > 0$.